Majorana Modes in the Machine

Circuit-Level Gate Noise on the VQE: Channels, Verification, and the Depth Optimum

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From a Validated VQE to a Noisy Device

What we already have

- A validated parity-constrained VQE: optimal $\theta^*(\mu)$ across the phase diagram, benchmarked against exact diagonalization.
- A noiseless depth diagnostic: subspace infidelity *falls* as the ansatz repetitions r grow.
- Conclusion so far: a deeper ansatz prepares a better state.

The question now

- Put *real gate noise* into that preparation circuit.
- Each repetition adds an entangling layer; every CNOT carries an error channel.
- As the circuit deepens, does the edge-string diagnostic survive?

Same Hamiltonian and edge string $O_{\mathrm{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$ as before; the new ingredient is **noise that enters through the gates**.

Gate Errors as Quantum Channels

A noisy gate is the ideal unitary followed by a CPTP error channel:

$$\tilde{\mathcal{U}}_g = \mathcal{E}_g \circ \mathcal{U}_g, \qquad \mathcal{E}_g(\rho) = \sum_k K_k \, \rho \, K_k^\dagger, \qquad \sum_k K_k^\dagger K_k = I. \tag{1}$$

A full circuit composes these gate-by-gate:

$$\rho_{\text{out}} = \left(\mathcal{E}_{g_M} \circ \mathcal{U}_{g_M}\right) \circ \cdots \circ \left(\mathcal{E}_{g_1} \circ \mathcal{U}_{g_1}\right)(\rho_{\text{in}}). \tag{2}$$

Key consequence

Noise enters *through the gates*: the number of error channels grows with the gate count, hence with the circuit depth. A density-matrix simulation with a gate-level `NoiseModel` captures this exactly.

The Channels, and Which Ones Respect Parity

Integrating the Lindblad dynamics over a gate duration τ turns physical decoherence into CPTP channels. What matters for the Majorana diagnostic is how each acts on parity $\hat{P} = \prod_j Z_j$.

Channel	Physical origin	Effect on the diagnostic
Depolarizing	isotropic, symmetric noise	shrinks all correlations / string contrast
Amplitude damping (T_1)	relaxation $\|1\rangle \to \|0\rangle$	parity-violating: leaks into the odd sector
Phase damping (T_2)	loss of phase coherence	parity-preserving: kills endpoint coherence
Thermal relaxation	T_1 and T_2 combined	the realistic device channel
Readout	bit-flip at measurement	biases the ± 1 parity-product estimator

Two effects are qualitatively distinct: T_2 erodes the *contrast* of the non-local string, while T_1 *poisons the parity sector* the diagnostic lives in.

Which figure uses which: depth study (slides 5/8/9) uses **depolarizing**; verification (slide 7) compares **depolarizing vs. thermal**; parity study (slides 11/12) uses single-qubit **phase/amplitude/depolarizing** on the frozen state.

Noise Integrated Through the Gates, by Depth

The linear `EfficientSU2` ansatz at depth r has

$$\#\,\text{CNOTs} = r\,(L-1),$$

and each `cx` carries a **depolarizing** channel (circuit-level), so the measured signal factorizes into two opposing factors:

$$|\langle O_{\text{edge}}\rangle|_{\text{meas}}(r) \;\approx\; \underbrace{|\langle O_{\text{edge}}\rangle|_{\text{ideal}}(r)}_{\text{expressibility}} \;\times\; \underbrace{(1-p_{\text{cx}})^{\,r(L-1)}}_{\text{accumulated gate noise}\,\downarrow} \quad . \tag{3}$$

- Expressibility has a **threshold**: below a minimum depth the ansatz cannot represent the state ($|O|_{\text{ideal}} \approx 0$); above it, $|O|_{\text{ideal}}$ saturates near 1.
- Gate noise then *falls* exponentially in depth.

The depth trade-off

The optimum is the **shallowest adequately-expressive depth**: just deep enough to prepare the state, no deeper, because every extra layer only adds CNOT noise.

From State-Vector VQE to a Noisy Density Matrix

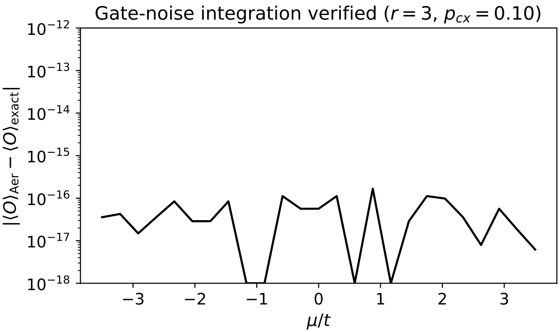
Earlier blocks read $\langle O_{\text{edge}} \rangle$ from an exact *state vector*; why use Aer's *density-matrix* simulator now?

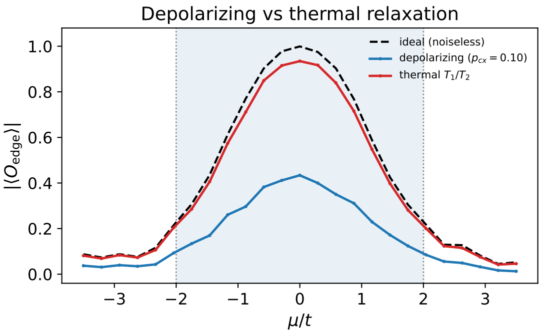
- **Same VQE, same circuit.** The noisy run uses the identical validated ansatz and optimal angles $\theta^*(\mu)$ from Block 3 – *the VQE is not replaced*; only how we evolve and read the state changes.
- **What changes.** A gate-level `NoiseModel` is attached, and $\langle O_{\text{edge}} \rangle$ is read as $\mathrm{Tr}[O_{\text{edge}}\, \rho]$ from a density matrix instead of from an ideal state vector.
- **Why a density matrix is required.** Noise turns the pure state *mixed*; a state vector cannot represent mixedness, so the density-matrix formalism is mandatory once gates are noisy. Aer evolves ρ through the circuit under the `NoiseModel`.
- **Why keep θ^* fixed** (no re-optimization under noise): to *isolate the variable* – the state is already good, so any signal loss is due to noise alone. (Noisy VQE is deferred; see "Next".)

Also: the channel fires after *every* `cx` (it accumulates with depth); reading $\mathrm{Tr}[O_{\text{edge}}\rho]$ avoids shot noise.

Precise Verification + a Physical T_1/T_2 Model

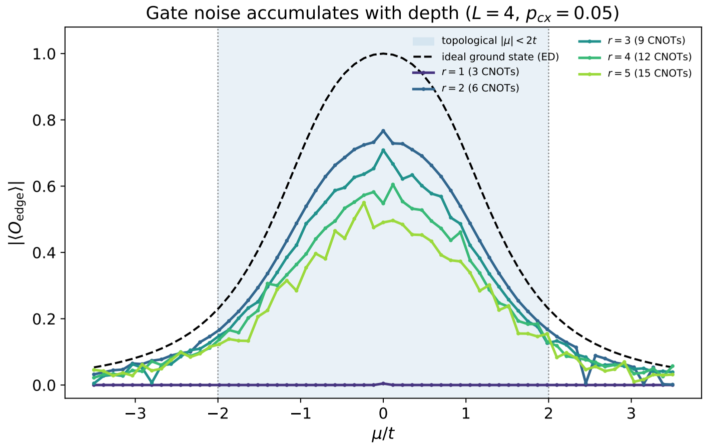
Verified circuit-level gate noise and a physical T_1/T_2 model





Left: the Aer `NoiseModel` edge string vs. an *independent* exact gate-by-gate density-matrix evolution (same circuit, depolarizing `SuperOp` after each `cx`) agree to $\sim 10^{-15}$ across all μ – the gate noise is correctly integrated through the whole circuit. **Right:** swapping depolarizing for `thermal_relaxation_error`(T_1, T_2, τ) gives the realistic channel; T_1 additionally poisons the parity sector, T_2 kills the endpoint coherence the edge string needs.

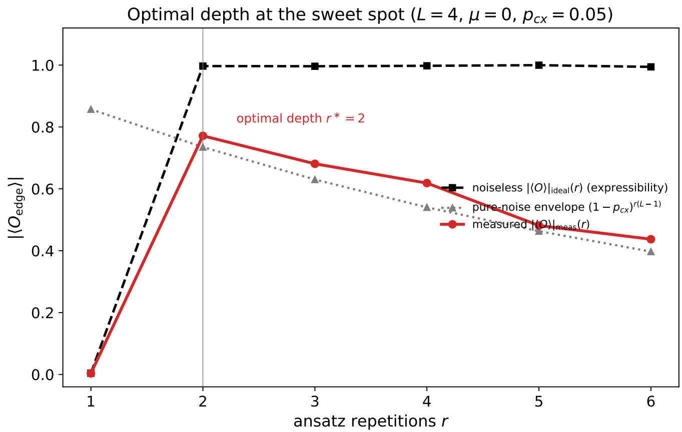
Gate Noise vs. Depth Across the Phase Diagram



Per-cx **depolarizing** channel at fixed strength p_{cx} (circuit-level); only the CNOT count $r(L-1)$ changes between curves.

- $r = 1$ is *under-expressive*: the diagnostic is dead everywhere (flat at zero).
- $r = 2$ first reaches the state, then deeper circuits are pushed *down* by accumulated CNOT noise.
- The phase contrast is sharpest at the $\mu = 0$ sweet spot and shallow depth.

The Depth Optimum, Made Explicit

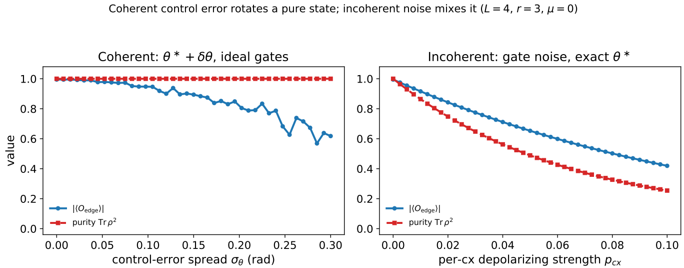


Channel: per-cx **depolarizing** (circuit-level); the envelope $(1 - p_{\text{cx}})^{r(L-1)}$ is the depolarizing attenuation. At the $\mu = 0$ sweet spot, the two factors separate:

- **Expressibility** (dashed): a step – zero at $r = 1$, saturated for $r \geq 2$.
- **Pure-noise envelope** (dotted): smooth $(1 - p_{\text{cx}})^{r(L-1)}$ decay.
- **Measured** (solid): $\approx$ their product, peaks at $r^* = 2$.

The optimum is not a smooth balance – it is the *first* depth that can express the state.

Two Families of Error: Coherent vs. Incoherent

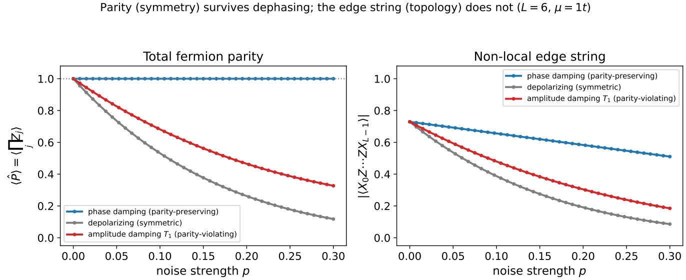


Same prepared state, two error types:

- **Coherent** $(\theta^* + \delta\theta)$: miscalibrated angles, **ideal (noiseless) gates** (no channel). The state stays *pure* ($\mathrm{Tr}\,\rho^2 = 1$) – only rotated off target.
- **Incoherent** (per-cx **depolarizing** channel, circuit-level): genuine decoherence – signal *and* purity fall.

A coherent error can be re-absorbed by recalibration or the VQE; incoherent loss cannot.

Professor's Q1: Is the Parity Protected by Topology?

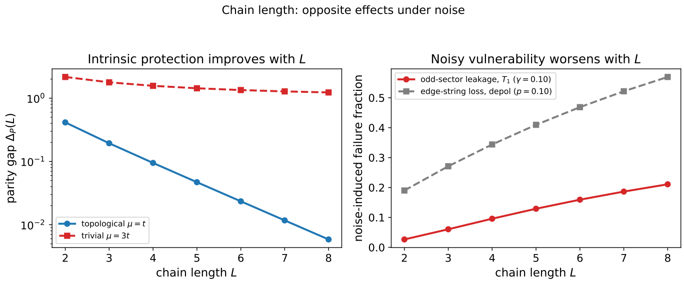


Topological state ($L = 6$, $\mu = t$). Channels applied as **single-qubit** maps on the exact even-parity state (*frozen-state*, not circuit-level): **phase damping**, **depolarizing**, **amplitude damping (T_1)**.

- **Left:** $\langle \hat{P} \rangle$ stays *exactly* 1 under phase damping – but that is the $\mathbb{Z}_2$ *symmetry* (true in every phase), **not** topology. It collapses under depolarizing, leaks under T_1.
- **Right:** the topological edge string decays under *every* channel – even the parity-preserving one.

Answer: parity is symmetry-protected; the topological signal is not noise-immune at fixed L.

Professor's Q2: How Does the Chain Length Matter?



Length acts in *opposite* directions. Right panel uses **single-qubit amplitude damping (T_1)** and **single-qubit depolarizing** on the **frozen** ground state at fixed strength γ.

- **Left:** the intrinsic parity gap is protected exponentially, $\Delta_P(L) \sim e^{-L/\xi}$ – longer chains are intrinsically *better*.
- **Right:** at fixed noise, T_1 odd-sector leakage and depolarizing edge-string loss both *grow* with L – longer chains are *noisier* to operate.

Answer: the two trends oppose, defining an optimal working length – the chain-length analogue of the optimal depth.

Summary and What This Enables Next

- Gate noise is integrated at the *circuit* level via a Qiskit `NoiseModel`, verified to machine precision against an exact gate-by-gate reference.
- The diagnostic is bounded by an expressibility threshold, then eroded by accumulated CNOT noise: the optimum is the **shallowest adequately-expressive depth**.
- Parity is protected by *symmetry*, not topology; the topological signal is not noise-immune at fixed L, and longer chains help the gap but hurt the device.

Next

- **Noisy VQE optimization**: a `NoiseModel`-backed estimator, so every cost evaluation is corrupted.
- **Backend-calibrated noise**: `NoiseModel.from_backend(...)` for real T_1/T_2 and gate-error data.

Code: `src/block4.py` (plots 2–7).
Notes: `week9_note.tex`, `parity_topology_protection.tex`.